TREND TESTS

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CEVE 543

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NULL HYPOTHESIS SIGNIFICANCE TESTING

1. State a **null hypothesis** H_0, such as “no trend”
2. Choose a **test statistic**
3. Work out (derive or simulate) the statistic’s distribution **if** H_0 **were true**
4. Compare the observed statistic to that distribution

The **p-value** is the probability, if H_0 were true, of a statistic at least as extreme as the one observed.

- A small p-value means data like these would be unusual under H_0.
- Reject H_0 when the p-value is below a level chosen in advance, usually 0.05.

WHAT A P-VALUE IS NOT

- A p-value is $P(\text{data this extreme} \mid H_0)$.
- It is **not** $P(H_0 \mid \text{data})$, such as the probability that there is no trend (Serinaldi et al., 2018, p. 136).
- A large p-value does not show that H_0 is true.

SIDE RANT

A p-value is useful when interpreted carefully. The trouble comes from treating a hypothesis as simply true or false, and sorting effects into “real” and “not real”. For more, read:

- McShane et al. (2017)
- Gelman & Hennig (2017)
- Stark & Saltelli (2018)
- Gelman & Shalizi (2013)

THE REGRESSION SLOPE TEST

- Fit $y_t = \beta_0 + \beta_1 t + \varepsilon_t$ by least squares; test $\beta_1 = 0$ with a t statistic.
- Assumes independent, normal residuals with constant variance.

THE MANN-KENDALL TEST

- $S = \sum_{i<j} \text{sign}(y_j - y_i)$: pairs where the later value is larger, minus pairs where it is smaller.
- Under H_0 (independent, identically distributed years, no ties), S is approximately normal with mean 0 and variance $n(n-1)(2n+5)/18$.
- Uses only the ordering, so it assumes no distribution for y.

TREND TESTS ON RECORDS WE HAVE SEEN

Record	Years	Regression p	Mann-Kendall p
Hobby rainfall	1940-2025	0.13	0.19
Brays Bayou peak flow	1929-2024	< 0.001	< 0.001
Sewells Point water level	1928-2026	< 0.001	< 0.001
… minus mean sea level	1928-2026	0.64	0.97
Galveston water level	1904-2025	< 0.001	< 0.001
… minus mean sea level	1904-2025	0.60	0.41

"… minus mean sea level" subtracts each year's mean water level before taking the maximum, as in week 5. For supplementary notes on both tests, read Helsel et al. (2020) §12.2, pp. 332-336.

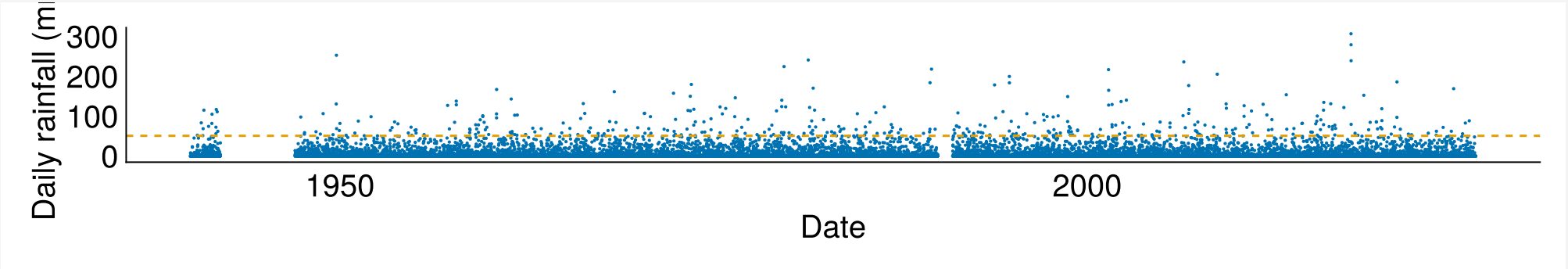
READING THE TABLE

- Both coastal records reject “no trend”, and neither does once mean sea level is subtracted.
- Brays Bayou rejects; Hobby does not.

Think

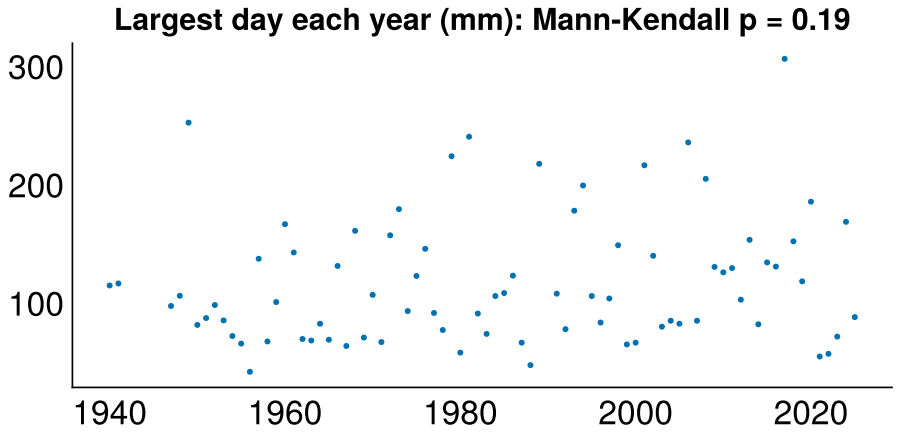
If we already know the cause, what did the test add?

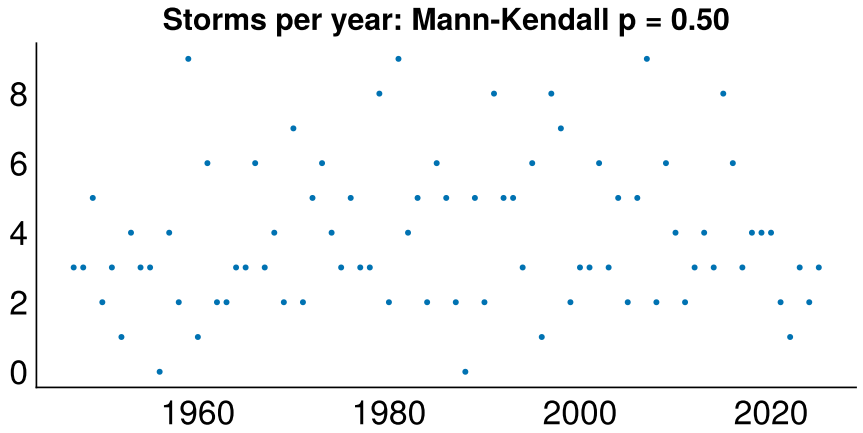
DAILY RAINFALL AT HOBBY

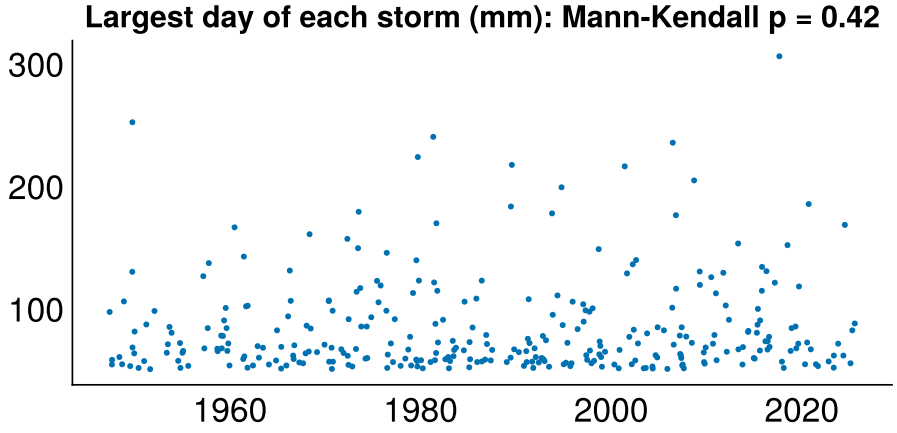


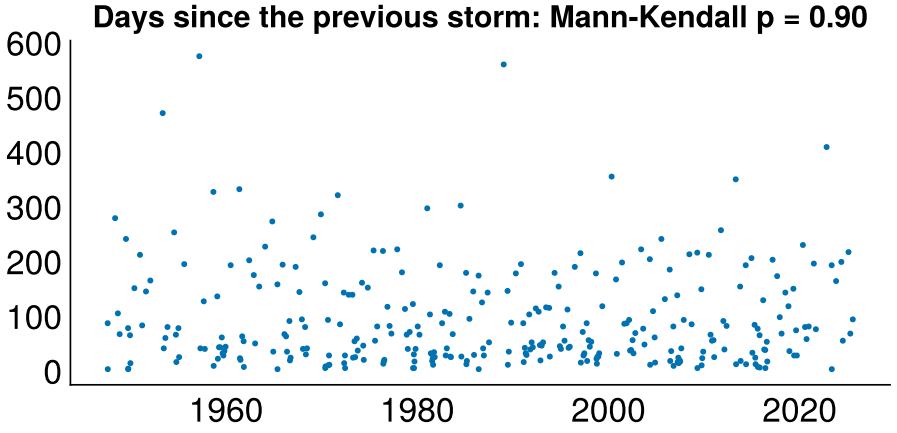
307 storms above 2 inches (dashed) from 1947 to 2025, counting rainy days within 3 days of each other as one storm.

FOUR THINGS THAT COULD BE CHANGING AT HOBBY









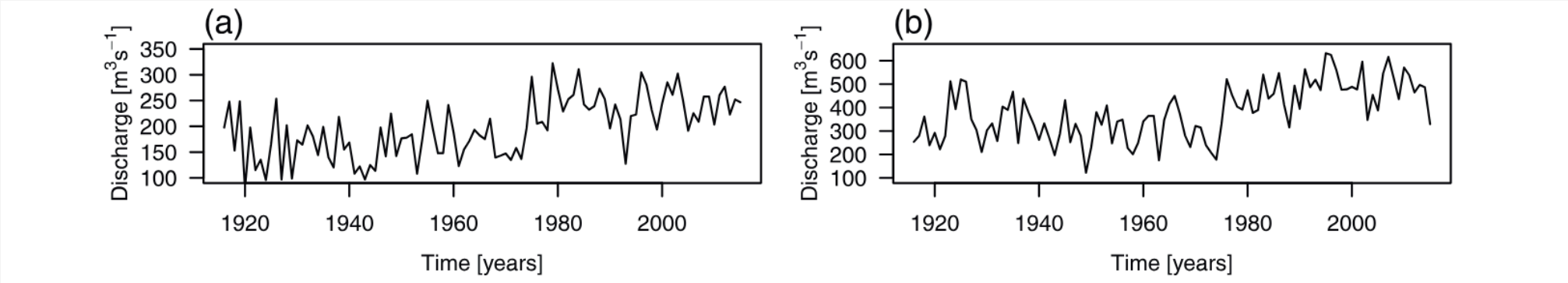
A SLOPE CAN BE LARGE AND NOT SIGNIFICANT

- Hobby's annual maxima rise 0.39 mm per year, with p = 0.13.
- That is 33 mm over 85 years, 28% of the average annual maximum.

Think

What do you think is changing at Hobby, and why?

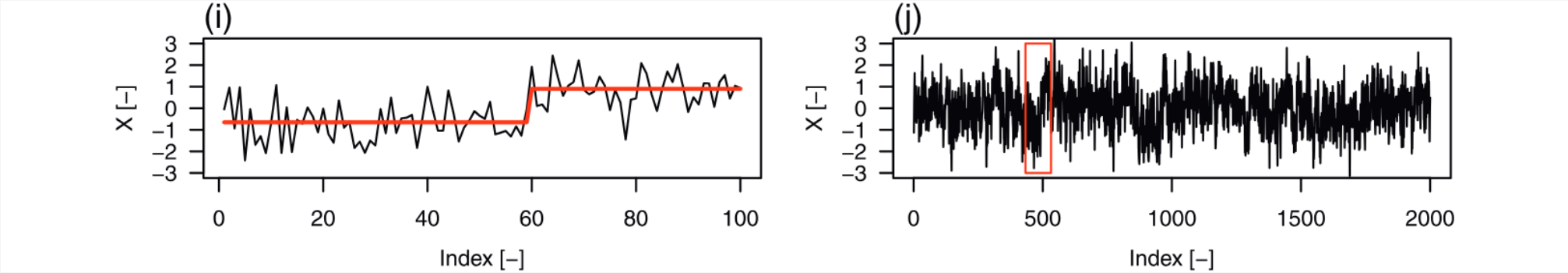
TWO NEARBY RIVERS WITH A STEP CHANGE AROUND 1975



Average annual discharge, 1916-2015, for two rivers the authors call the Nera and the Velino (Serinaldi et al., 2018, Fig. 1a-b, CC BY 4.0).

Every change-point test rejects “no change” for both (Serinaldi et al., 2018, p. 133).

NEITHER SERIES IS A RIVER



The “Nera” is independent noise plus a step; the “Velino” is a window, in the red box, of a stationary process with long memory (Serinaldi et al., 2018, Fig. 1i-j, CC BY 4.0).

MEMORY ALONE MAKES MANN-KENDALL REJECT

Annual maximum flow at 250 US streamflow gauges (Serinaldi et al., 2018, pp. 147-148):

- standard Mann-Kendall rejects “no trend” at 46;
- a null that keeps each river’s memory rejects at 12;
- chance alone, at the 5% level, gives about 12.